

BINOY ANIL

Software Engineer | Full-Stack, ML & Systems

[linkedin.com/in/binoy-anil-641474230](https://www.linkedin.com/in/binoy-anil-641474230) | github.com/bino369 | binofolio.vercel.app

Summary

Versatile Software Engineer with hands-on expertise spanning full-stack web development, machine learning, and systems programming. Proven track record of building production-grade software—from deep learning computer vision pipelines and real-time AirPlay media receivers to scalable full-stack web platforms. Passionate about engineering high-performance systems and solving complex technical challenges.

Education

Bachelor of Computer Applications (BCA)

St. Berchmans College Autonomous, Changanacherry

2023 – Present

Coursework & CS Foundations

- Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java), DBMS, Operating Systems, Computer Networks.
- Actively applying academic computer science theory to open-source software engineering and real-world system design.

Projects

EcoIdentify-AI

PyTorch, EfficientNetV2, FastAPI, React

github.com/Bino369/EcoIdentify-AI

- Built an end-to-end waste classification web application using PyTorch and EfficientNetV2, achieving high multi-class inference accuracy.
- Implemented real-time image preprocessing and inference pipeline with FastAPI backend and responsive React frontend for live webcam classification.

Live-Emotion-Detection

Python, CNN, FER2013, FastAPI, React

github.com/Bino369/Live-Emotion-Detection

- Developed a real-time facial emotion recognition system using a CNN trained on the FER2013 dataset, achieving low-latency affect classification.
- Engineered a lightweight FastAPI streaming API integrated with a React UI for 30+ FPS live webcam feed analysis and instant sentiment visualization.

MirrorShare

C++20, Qt6, RTSP/RTP, Networking

github.com/Bino369/MirrorShare

- Built a native Windows AirPlay receiver as a high-performance alternative to UXPlay using modern C++20 and Qt6 for zero-lag screen mirroring and audio streaming.
- Implemented multi-threaded network packet handling, RTSP/RTP protocol decoding, and hardware-accelerated video rendering for stable low-latency playback.

BCA Attendance Tracker

TypeScript, React, Vite, Node.js, Express, MongoDB

github.com/Bino369/bca-attendance-tracker

- Built a full-stack attendance management platform with TypeScript React (Vite) and Node/Express, streamlining daily record keeping for faculty.
- Designed RESTful API endpoints with MongoDB/Mongoose for student analytics, attendance percentage calculation, and automated CSV export generation.

Certifications

Skill Advancement cum Project Mentorship Program

Indian Institute of Information Technology (IIIT) Kottayam

May 2026

Certificate ID: IIITK/SAPM-SBC/2026/ECERT/068

- Completed intensive machine learning mentorship under Dr. Sivaiah Bellamkonda, focusing on computer vision and practical ML deployments.
- Awarded certificate of completion certified by Dr. Radhakrishnan M, Registrar, IIIT Kottayam.

Skills

- **Languages:** C++ (C++20), Python, TypeScript, JavaScript, Core Java, HTML5, CSS3, SQL
- **Frameworks & Web:** React, Vite, Node.js, Express, FastAPI, Qt6, RESTful APIs, WebRTC
- **Machine Learning & CV:** PyTorch, Computer Vision, EfficientNetV2, CNNs, FER2013, OpenCV
- **Tools & Platforms:** Git, GitHub, MongoDB (Mongoose), Linux, CMake, VS Code, Postman